

Causal Inference and ML: Week 10

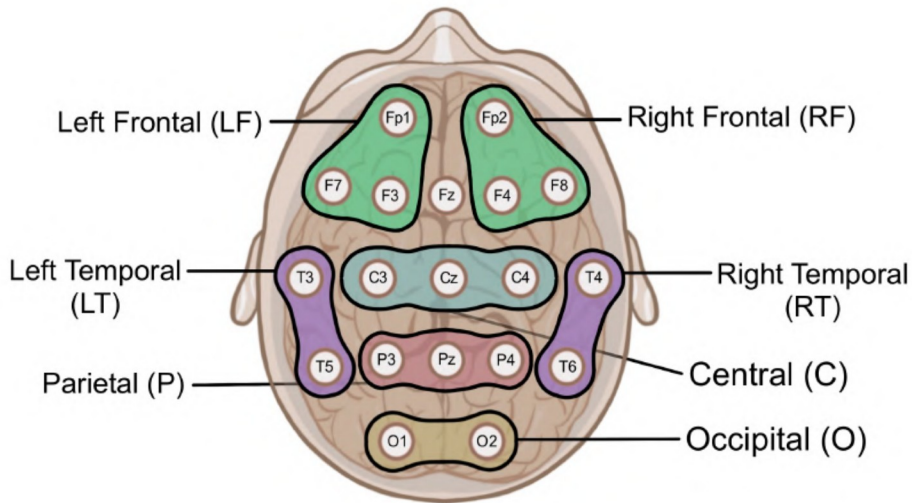
Causal Discovery in Time Series

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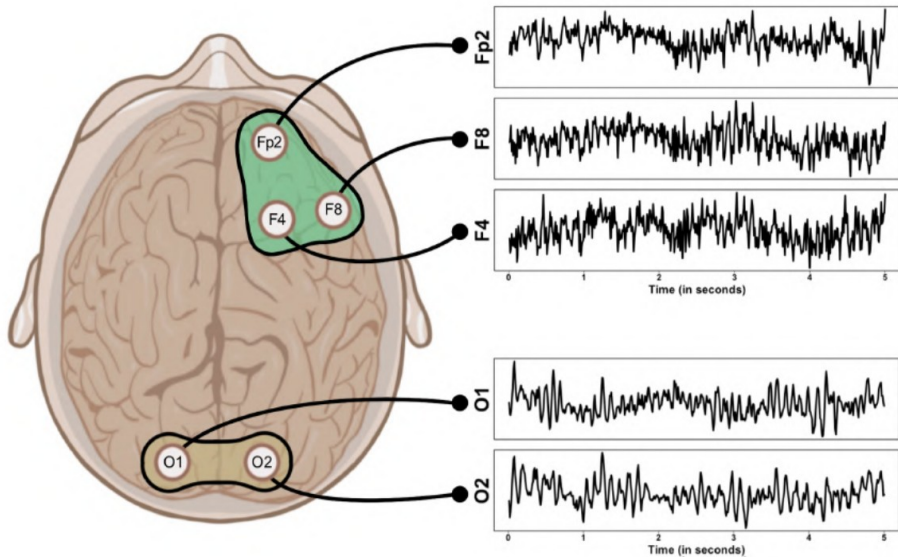
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Why Time Series Causal Discovery?

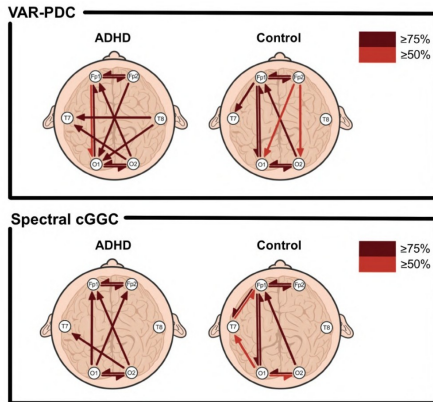
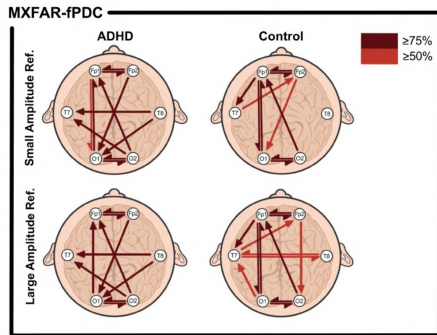
Why Time Series Causal Discovery?



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Why Time Series Causal Discovery?

Main idea

Multivariate time series appear in finance, climate science, neuroscience, and sensors. The challenge is to discover directed causal links between variables across time.

Why standard causal discovery is not enough

Time series are not i.i.d. and they exhibit:

- ▶ autocorrelation,
- ▶ lagged causal effects,
- ▶ contemporaneous dependence,
- ▶ and often non-stationarity.

Challenge

Infer causal links of the form $X_{t-\tau}^i \rightarrow X_t^j$ from observed multivariate time series.

Time Series Graphs

Definition

A time series graph consists of nodes X_t^i (variable i at time t) and directed edges representing causal dependencies.

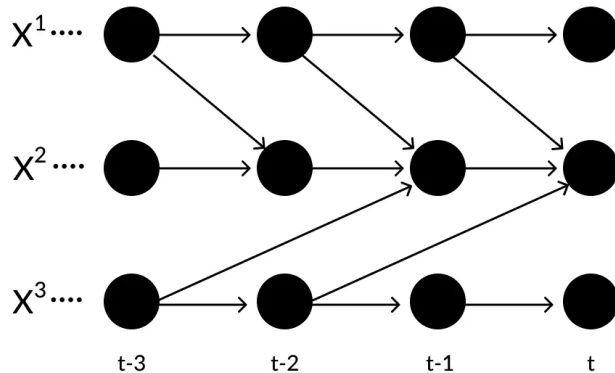
Link Types

- ▶ **Autoregressive links:** $X_{t-\tau}^i \rightarrow X_t^i$
- ▶ **Lagged links:** $X_{t-\tau}^i \rightarrow X_t^j$ ($\tau \geq 1$)
- ▶ **Contemporaneous links:** $X_t^i \rightarrow X_t^j$

Stationarity and Summary Graphs

Under stationarity, the same causal mechanism repeats over time. A *summary graph* aggregates these repeated lagged relationships into a variable-level causal representation.

Example: Summary Graph



Interpretation

Each arrow represents a causal influence that can repeat over time under the stationarity assumption.

Why FullCI Is Hard

FullCI idea

Full conditional independence tests condition on the full past of all variables.

Problem

This can become computationally expensive and statistically unstable when the number of variables or lags is large.

Solution

PCMCI reduces the conditioning set first, then tests candidate links more carefully.

PCMCI in Two Steps

Step 1: Condition selection

Use a PC-style algorithm to select a small set of candidate parents for each variable.

Step 2: Momentary Conditional Independence

Test each candidate link while conditioning on the selected parents of the source and target variables.

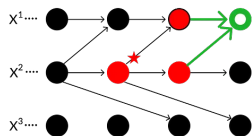
Benefit

PCMCI improves power and reduces false positives compared with conditioning on the full past.

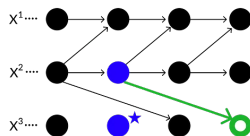
PCMCI in Two Steps

Phase 1 (PC)

PC result for parents of X^1

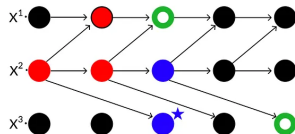


PC result for parents of X^3



Phase 2 (MCI)

MCI test $X^1 \xrightarrow{t-2} X^3$



Test:

$$X_{t-\tau}^i \perp\!\!\!\perp X_t^j | \hat{P}(X_t^j) \setminus \{X_{t-\tau}^i\}, \hat{P}(X_{t-\tau}^i)$$

FullCI Versus PCMCI

FullCI

Condition on the full past of all variables.

- ▶ conceptually simple,
- ▶ but computationally expensive,
- ▶ and often statistically unstable in high dimensions.

PCMCI

- ▶ first selects candidate parents,
- ▶ then tests links using MCI,
- ▶ and is more scalable for multivariate time series.

Main message

PCMCI keeps the useful parts of conditional independence testing while avoiding unnecessary conditioning.

Phase 1: Parent Selection

Idea

For each target variable, PCMCI searches for a compact set of candidate parents using iterative conditional independence tests.

Why it helps

- ▶ reduces the size of conditioning sets,
- ▶ improves power,
- ▶ and avoids overconditioning on irrelevant variables.

Output

The result is a superset of the true parents, which is then refined in the MCI step.

Phase 2: Momentary Conditional Independence

MCI idea

For a candidate link $X_{t-\tau}^i \rightarrow X_t^j$, test conditional dependence while controlling for the selected parents of both the source and the target.

Why this is important

This accounts for autocorrelation and indirect paths more effectively than a naive pairwise test.

Interpretation

If the link remains significant after conditioning, it is retained as a causal candidate.

PCMCI+

PCMCI+ extends PCMCI

PCMCI+ is designed for settings where contemporaneous links also matter.

Output

It can return a graph that includes:

- ▶ lagged causal links,
- ▶ contemporaneous relations,
- ▶ and endpoint marks that reflect uncertainty or hidden structure.

Why it matters

PCMCI+ is useful when same-time interactions cannot be ignored.

Getting Started with Tigramite

1. Install Tigramite and its optional independence test dependencies

```
!pip install tigramite dcor scikit-learn statsmodels
```

2. Install graphviz system-level dependency for advanced causal graph rendering

```
!apt-get install -y graphviz libgraphviz-dev
```

```
!pip install pygraphviz
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from tigramite import data_processing as pp
```

```
from tigramite.toymodels import structural_causal_processes as toys
```

```
from tigramite import plotting as tp
```

```
from tigramite.pcmci import PCMCI
```

```
from tigramite.independence_tests.parcorr import ParCorr
```

```
from tigramite.independence_tests.gpdc import GPDC
```

```
from tigramite.independence_tests.cmiknn import CMiknn
```


Structural Causal Process

The generated multivariate time series follows the linear structural equations

$$X_0(t) = -0.7 X_0(t-1) - 0.2 X_1(t-1) + \varepsilon_0(t),$$

$$X_1(t) = -0.5 X_1(t-1) - 0.8 X_3(t-1) + \varepsilon_1(t),$$

$$X_2(t) = 0.35 X_2(t-1) - 0.25 X_1(t-2) - 0.6 X_3(t-3) + \varepsilon_2(t),$$

$$X_3(t) = 0.9 X_3(t-1) + \varepsilon_3(t).$$

True causal links:

- ▶ $X_0(t-1) \rightarrow X_0(t), \quad X_1(t-1) \rightarrow X_0(t)$
- ▶ $X_1(t-1) \rightarrow X_1(t), \quad X_3(t-1) \rightarrow X_1(t)$
- ▶ $X_2(t-1) \rightarrow X_2(t), \quad X_1(t-2) \rightarrow X_2(t), \quad X_3(t-3) \rightarrow X_2(t)$
- ▶ $X_3(t-1) \rightarrow X_3(t)$

Synthetic Data Example

Toy model

We generate a small linear multivariate time series with lagged causal links.

```
np.random.seed(42)

def lin_f(x):
    return x

links_coeffs = {
    0: [((0, -1), 0.7, lin_f), ((1, -1), -0.8, lin_f)],
    1: [((1, -1), 0.8, lin_f), ((3, -1), 0.8, lin_f)],
    2: [((2, -1), 0.5, lin_f), ((1, -2), 0.5, lin_f), ((3, -3), 0.6, lin_f)],
    3: [((3, -1), 0.4, lin_f)],
}

T = 1000
data, true_parents = toys.structural_causal_process(links_coeffs, T=T)

var_names = ['X0', 'X1', 'X2', 'X3']
df = pp.DataFrame(data, var_names=var_names)
```

Running PCMCI

PCMCI workflow

```
pcmci = PCMCI(  
    dataframe=df,  
    cond_ind_test=ParCorr(),  
    verbosity=1  
)  
  
results = pcmci.run_pcmci(  
    tau_min=1,  
    tau_max=3,  
    pc_alpha=0.05  
)  
  
pcmci.print_significant_links(  
    p_matrix=results['p_matrix'],  
    val_matrix=results['val_matrix'],  
    conf_matrix=results.get('conf_matrix', None),  
    alpha_level=0.05  
)
```

Conditional Independence Tests

Common choices

- ▶ **ParCorr**: partial correlation for linear-Gaussian data.
- ▶ **RobustParCorr**: more stable when outliers are present.
- ▶ **GPDC**: nonlinear, continuous dependencies.
- ▶ **CMIknn**: nonparametric conditional mutual information estimation.

Practical rule

Choose the CI test according to the expected data-generating mechanism.

Inspecting Lagged Dependencies

Lag function plot

```
correlations = pcmci.get_lagged_dependencies(  
    tau_max=20,  
    val_only=True  
)['val_matrix']  
  
tp.plot_lagfuncs(  
    val_matrix=correlations,  
    setup_args={  
        'var_names': var_names,  
        'x_base': 5,  
        'y_base': 0.5  
    }  
)
```

Why it helps

Lag functions help identify which links appear at short or long lags and guide model choice.

Plotting the Result

Time series graph

```
tp.plot_time_series_graph(  
    figsize=(6, 4),  
    val_matrix=results['val_matrix'],  
    graph=results['graph'],  
    var_names=var_names,  
    link_colorbar_label='MCI'  
)
```

Process causal graph

```
tp.plot_graph(  
    val_matrix=results['val_matrix'],  
    graph=results['graph'],  
    var_names=var_names,  
    link_colorbar_label='cross-MCI',  
    node_colorbar_label='auto-MCI'  
)
```

How to Read the Output

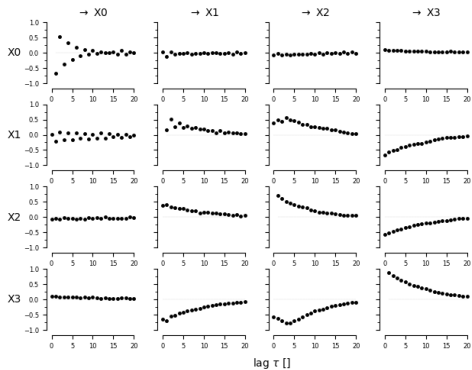
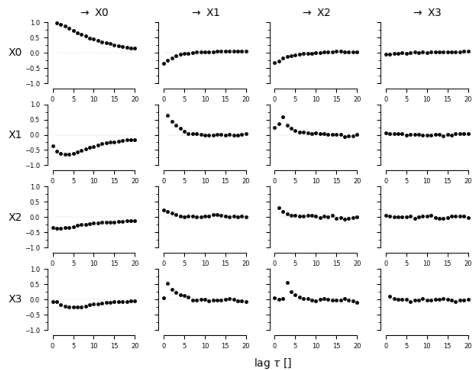
Three useful views

- ▶ time series graph,
- ▶ lag function plot,
- ▶ summary causal graph.

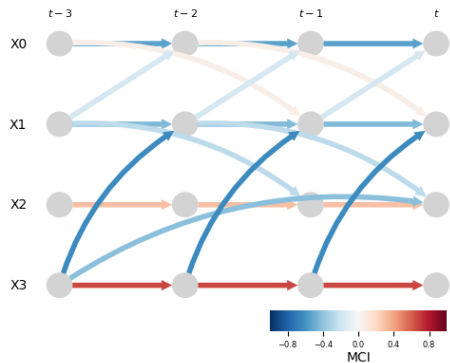
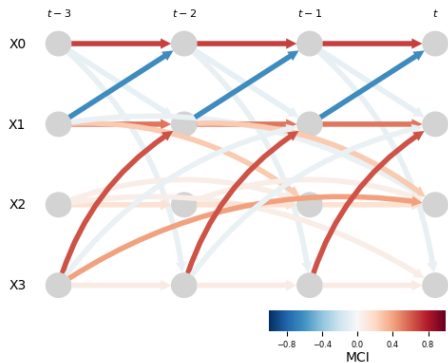
What to look for

- ▶ lag length,
- ▶ direction of the link,
- ▶ strength of dependence,
- ▶ and whether a pattern is consistent across time.

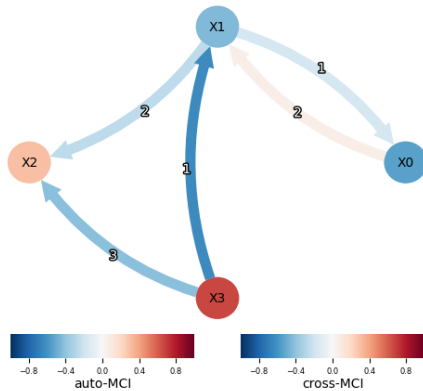
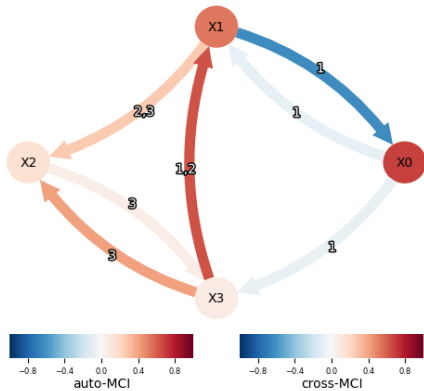
Inspecting Lagged Dependencies



Plotting the Result



Plotting the Result



Significance and Multiple Testing

Why correction is needed

PCMCI tests many links across variables and lags, so false positives can accumulate.

Common strategy

Use p-value correction or thresholding to control the false discovery rate.

Practical meaning

A link should be interpreted as robust only if it remains significant after correction.

When Basic PCMCI Is Not Enough

Harder settings

Real time series often contain contemporaneous links, latent confounders, and regime changes.

Helpful extensions

- ▶ **PCMCI+**: adds contemporaneous causal discovery.
- ▶ **LPCMCI**: allows latent confounding.
- ▶ **Regime-based extensions**: handle changing dynamics.

Takeaway

These methods extend the same causal-discovery logic to more realistic settings.

Practical Workflow in Tigramite

Recommended workflow

1. Inspect the data and choose variable names.
2. Visualize lagged dependencies.
3. Select a conditional independence test.
4. Run PCMCi with a reasonable lag window.
5. Inspect significant links and graphs.
6. Refine with PCMCi+ or an extension if needed.

Final message

Tigramite turns multivariate time-series causal discovery into a clear and reproducible Python workflow. But there are many alternatives for example:
CausalFlow - <https://github.com/lcastri/causalflow>

Key Takeaways

- ▶ Multivariate time series require lag-aware causal methods.
- ▶ PCMCI reduces the conditioning burden before testing links.
- ▶ Tigramite provides a practical Python workflow for causal discovery.
- ▶ CI test choice matters: ParCorr, GPDC, and CMIknn serve different data types.
- ▶ PCMCI+ and other extensions broaden the scope beyond basic lagged discovery.

Thank You!

Thanks to all participants for your engagement, enthusiasm, and valuable contributions throughout this course.

I hope the discussions we had, the questions you made and the given answers - made this a rewarding learning experience for everyone.

I wish you continued success in your future endeavors.

Thank you for being part of this journey.